

LETTER

Uptake of *Prochlorococcus*-derived metabolites by *Alteromonas macleodii* MIT1002 shows high levels of substrate specificity

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Scientific Significance Statement

Microbes recycle carbon in the ocean, converting carbon dioxide into dissolved organic matter and back again. This interconversion represents a substantial carbon flux, making it an important component of global carbon cycling. Dissolved organic matter is a complex mix of thousands of molecules, but it is typically considered in bulk. This leaves us without a nuanced understanding of how chemical composition affects microbial turnover of dissolved carbon. In this paper, we grew a microbe, *Alteromonas macleodii* MIT1002, with a series of specific molecules which we know are produced by a dominant phytoplankton, *Prochlorococcus*. We found that MIT1002 was surprisingly selective and only used a few specific molecules for growth, even among very similar compounds. We further showed that this specificity is likely a result of MIT1002's ability to take up molecules from its environment. Our results show that a more nuanced look at dissolved organic matter expands our understanding of both carbon cycling and microbial ecology in the ocean.

Abstract

Seawater contains small biomolecules, or metabolites, that are highly labile components of dissolved organic matter (DOM). Marine microbes interact by exchanging metabolites, thus shaping marine microbial ecology, DOM composition, and global carbon cycling. To better constrain one set of microbe-metabolite interactions, we cultured the marine gammaproteobacterium *Alteromonas macleodii* MIT1002 on a range of compounds excreted by a sympatric cyanobacterium, *Prochlorococcus*. *Alteromonas macleodii* MIT1002 could metabolize the branched-chain amino acids leucine, isoleucine, and valine, as well as 3-methyl-2-oxobutanoic acid (a branched-chain ketoacid intermediate of valine metabolism), but not thymidine, kynurenine, 4-hydroxybenzoic acid, nor the other branched-chain ketoacids. The utilization of branched-chain amino acids indicates that *A. macleodii* MIT1002 can metabolize each corresponding ketoacid, suggesting that transporter specificity underlies the observed substrate specificity for 3-methyl-2-oxobutanoic acid. These experiments show that even subtle changes in chemical structure can result in different microbial interactions and different fates for dissolved metabolites.

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Dissolved organic carbon (DOC) comprises 662 Pg C in the ocean, a reservoir comparable to atmospheric carbon dioxide (Hansell et al. 2009). Marine microbes influence DOC composition and flux, and molecules in this carbon pool mediate microbial interactions. Elucidating these microbe-molecule connections is therefore crucial to understanding both microbial ecology and global carbon cycling (Moran, Kujawinski, et al. 2022). While most studies have examined the response of microbes to bulk dissolved organic matter (DOM) or broad molecular subsets of DOM (Givati et al. 2024; Liu et al. 2020; Moran, Ferrer-González, et al. 2022; Pedler et al. 2014), the specificity of interactions between microbes and DOM molecules remains an open question. For example, recent work has shown that diverse heterotrophic bacteria preferentially use different classes of organic carbon in a way that cannot be predicted by phylogeny (Forchielli et al. 2022). Similarly, bacterial growth efficiency (BGE), or the fraction of carbon incorporated into biomass relative to carbon uptake, is expected to vary with both microbial taxa and organic substrate (Saifuddin et al. 2019). This heterogeneity implies that a change in either the microbial community or DOM composition could significantly impact the ecology and carbon flux of marine environments.

The picocyanobacterium *Prochlorococcus* is the dominant phototroph in subtropical oligotrophic oceans (Partensky et al. 1999), responsible for fixing an estimated $\sim 4\text{--}8$ Pg C per year (Flombaum et al. 2013; Letscher et al. 2023). An estimated 2–25% of this carbon is exuded as DOC by actively growing cells (Bertilsson et al. 2005; López-Sandoval et al. 2013). The quantity and composition of this exudate vary with strain and environmental conditions, providing a complex set of substrates underpinning interactions between *Prochlorococcus* and other marine microbes (Becker et al. 2019; Biller et al. 2016; Calfee et al. 2022; Kujawinski et al. 2023; Ofaim et al. 2021; Sher et al. 2011; Szul et al. 2019).

Alteromonas macleodii is a model marine copiotrophic gammaproteobacterium (Pedler Sherwood et al. 2015) often isolated from *Prochlorococcus* enrichments (Aharonovich and Sher 2016; Biller et al. 2015, 2016; Coe et al. 2016). This heterotrophic microorganism has been used to probe microbial sinks of DOC, as *A. macleodii* cultures can decrease bulk DOC concentrations as effectively as a complex microbial consortium (Pedler et al. 2014). In co-cultures, *A. macleodii* lives mutualistically with *Prochlorococcus* (Kearney et al. 2021), easing oxidative stress for the phototroph while relying on *Prochlorococcus*-derived DOC (Biller et al. 2016; Morris et al. 2008, 2011, 2012). Individual strains of *A. macleodii* use different subsets of organic molecules, perhaps reflecting the environments from which each strain was isolated (Ivars-Martinez et al. 2008; Koch et al. 2020; Manck et al. 2020). We hypothesized that *A. macleodii* strain MIT1002, which was isolated from a *Prochlorococcus* enrichment (Biller et al. 2015), should metabolize a wide range of substrates released by *Prochlorococcus*. Although recent work quantified

Prochlorococcus-derived metabolites (Kujawinski et al. 2023), the fate of those molecules and their effect on microbial ecology remains unconstrained. Here we investigate the response of *A. macleodii* MIT1002 to *Prochlorococcus*-derived metabolites and uncover evidence for high degrees of substrate specificity.

Methods

Strain and media

All experiments were conducted with *A. macleodii* MIT1002, isolated from *Prochlorococcus* NATL2A cultures (Biller et al. 2015). Under sterile conditions, cultures were revived from cryopreserved glycerol stocks and grown in ProMM media (Berube et al. 2015) (Supporting Information). Cultures were monitored by OD₆₀₀ and transferred three times during mid-exponential growth before each experiment. Before uptake experiments, cultures were quantified by flow cytometry and transferred during mid-exponential growth when cell abundances reached between 7×10^7 and 2×10^8 cells mL⁻¹. Both ProMM and experimental media were prepared in filtered (0.2 μ m, Pall PES membrane filters), autoclaved, Vineyard Sound seawater (Supporting Information).

Growth experiments

Preliminary experiments confirmed that *A. macleodii* MIT1002 could grow without exogenous vitamins, on pyruvate alone, and on lower carbon concentrations than present in ProMM (Supporting Information Figs. S1, S2). To test the assimilation of metabolites known to be exuded by *Prochlorococcus*, we cultured *A. macleodii* MIT1002 on a series of these compounds. In each growth experiment, *A. macleodii* MIT1002 received either a *Prochlorococcus*-associated metabolite, pyruvate, or a mix of pyruvate and the metabolite. Pyruvate acted as a positive control. Mixes of pyruvate and the selected *Prochlorococcus* metabolite were tested, at multiple pyruvate concentrations, in case metabolite utilization was enabled when pyruvate supplemented the energetic demands of *A. macleodii* MIT1002. Metabolites tested were present in *Prochlorococcus* spent media (Kujawinski et al. 2023) and included: thymidine; 4-hydroxybenzoic acid; kynurenine; the branched-chain amino acids valine, isoleucine, and leucine; and the branched-chain ketoacids 3-methyl-2-oxobutanoic acid (intermediate of valine metabolism), 3-methyl-2-oxopentanoic acid (intermediate of isoleucine metabolism), and 4-methyl-2-oxopentanoic acid (intermediate of leucine metabolism). Cultures were grown with 0 or 900 μ M C from the *Prochlorococcus*-derived metabolite, and 0, 900, or 1800 μ M C from pyruvate (totaling 0, 900, 1800, or 2700 μ M C). For reference, ProMM contains 6900 μ M C and marine organic carbon concentrations typically range from 40 to 80 μ M C (Hansell et al. 2009). If a metabolite failed to support *A. macleodii* MIT1002 growth, results were verified with repeat experiments (Supporting Information Figs. S3–S5).

Growth experiments were conducted on black walled, clear bottom, 96-well plates using a BioTek Synergy 2 plate reader,

measuring OD₆₀₀ every 30 min for 48 h. Milli-Q blanks were placed in wells along the border of the 96-well plate to mitigate growth discrepancies due to evaporation in exterior wells. Additional growth discrepancies due to well position were occasionally observed on the next row or column in from the edge. If observed, the entire row or column with a discrepancy was removed from analysis for that experiment. On average, two edge rows or columns were removed per experiment. Experiments were conducted with six replicates for each condition. After removing well plate columns or rows where replicates crashed, we were left with at least three replicates for each condition (except for the leucine growth experiment, where the 900 μ M C from pyruvate condition was left with two replicates). The average signal in the 0 μ M C negative controls was subtracted from each well as a baseline correction, to account for trace seawater carbon. The carrying capacity in each well was calculated as the difference between the maximum and minimum OD₆₀₀ observed in that well. The significance of the effect of carbon substrates on carrying capacity was calculated using one-way ANOVA with a post-hoc Tukey's HSD test.

Uptake experiments

To test whether *A. macleodii* MIT1002 could take up compounds without a concomitant increase in biomass, we measured the uptake of 3-methyl-2-oxobutanoic acid and 3-methyl-2-oxopentanoic acid during exponential growth by monitoring the change in media (i.e., extracellular) metabolite concentrations. Before uptake experiments, cultures were washed 3 times with unamended sterile seawater by centrifugation for 10 min at 7000 $\times$ g. Cultures were then grown in 100 mL of minimal media containing seawater, inorganic nutrients, and 180 μ M of a selected metabolite (Supporting Information) in 125 mL acid-washed polycarbonate bottles. Control cultures were conducted with no metabolite added. One culture was grown in parallel with experimental cultures, using 3-methyl-2-oxobutanoic acid as the carbon source, and sampled repeatedly to track growth and guide sampling. At each timepoint, cultures and controls were sampled destructively in triplicate. 750 μ L of each culture or control was fixed with paraformaldehyde and stored at -20°C until analysis by flow cytometry (Supporting Information Methods). The remaining culture was filtered through an Omnipore (Millipore) 0.2 μ m PTFE membrane filter. Filtrate was acidified to pH $\sim$ 3 using concentrated hydrochloric acid and stored at 4°C for less than 48 h until PPL extraction.

Acidified filtrate was extracted with 100 mg, 1 mL Bond Elut PPL cartridges (Agilent, Santa Clara, CA, USA) as previously described (Dittmar et al. 2008; Kido Soule et al. 2015). Extracts were dried on a vacufuge (Eppendorf) and reconstituted in Milli-Q water with deuterated biotin (final concentration 0.05 mg mL⁻¹) and ¹³C₅-3-methyl-2-oxobutanoic acid (final concentration 0.02 μ g mL⁻¹). Extracts were stored at -20°C until analysis by liquid chromatography–tandem

mass spectrometry (LC–MS/MS; Supporting Information Methods).

Standard curves were constructed using at least a five-point matrix-matched standard curve which spanned the range of observed sample signal intensities. For 3-methyl-2-oxobutanoic acid, standard curves were constructed using the signal of unlabeled 3-methyl-2-oxobutanoic acid normalized to the signal of ¹³C₅-3-methyl-2-oxobutanoic acid. 3-Methyl-2-oxopentanoic acid standard curves were constructed using the signal of unlabeled 3-methyl-2-oxopentanoic acid without additional normalization. Normalization to a stable isotope labeled internal standard improved the precision of quantification, but did not change the measured values substantially. The use of isotope labeled internal standards is often restricted for practical reasons to metabolites where precise quantification is especially important. We therefore used an internal standard for 3-methyl-2-oxobutanoic acid, which we hypothesized would change in concentration over time, but not 3-methyl-2-oxopentanoic acid, which we hypothesized would stay constant. Measured concentrations were corrected for extraction efficiency (Johnson et al. 2017; Kujawinski et al. 2023). For 3-methyl-2-oxobutanoic acid, metabolite concentrations were normalized to cell

Table 1. Metabolites tested for their ability to support *A. macleodii* MIT1002 growth. *A. macleodii* MIT1002 was grown with either one of the listed metabolites (all known to be part of *Prochlorococcus* exudates), pyruvate, or a mix of pyruvate and the listed metabolite. A metabolite was considered to support *A. macleodii* MIT1002 growth if carrying capacity was significantly higher when *A. macleodii* MIT1002 cultures received the metabolite than when *A. macleodii* MIT1002 received no *Prochlorococcus* metabolite. BCAA, branched chain amino acid.

Metabolite	Supported <i>A. macleodii</i> MIT1002 growth?	Biochemical role
Leucine	Yes	BCAA metabolism
4-Methyl-2-oxopentanoic acid	No	BCAA metabolism
Isoleucine	Yes	BCAA metabolism
3-Methyl-2-oxopentanoic acid	No	BCAA metabolism
Valine	Yes	BCAA metabolism
3-Methyl-2-oxobutanoic acid	Yes	BCAA and pantothenate metabolism
Kynurenine	No	Tryptophan metabolism
Thymidine	No	Pyrimidine metabolism
4-Hydroxybenzoic acid	No	Ubiquinone metabolism

concentrations. The resulting per-cell bulk uptake rate was treated as a pseudo-first order process and modeled as an exponential decay function where C_t is the concentration at time t , C_0 is the initial concentration, and k is the per-cell uptake rate (Eq. 1).

$$C_t = C_0 e^{-kt} \quad (1)$$

Results

Growth experiments

Of the metabolites tested, only the branched-chain amino acids (valine, isoleucine, and leucine) and the valine precursor 3-methyl-2-oxobutanoic acid supported *A. macleodii* MIT1002 growth (Table 1; Figs. 1, 2; Supporting Information Figs. S6–S8; Halloran et al. 2025). However, at constant carbon concentrations, carrying capacity was significantly higher on pyruvate alone than on any of these four metabolites, or on a mix of pyruvate and any of these metabolites (Fig. 2). In contrast, 3-methyl-2-oxopentanoic acid, 4-methyl-2-oxopentanoic acid, thymidine, kynurenine, and 4-hydroxybenzoic acid did not support growth (Figs. 1, 2; Supporting Information Figs. S3–S5).

Uptake experiments

To test whether *A. macleodii* MIT1002 is taking up metabolites even when they are not used for net growth, we conducted uptake experiments comparing two structurally and biochemically related metabolites: 3-methyl-2-oxobutanoic

acid, the valine precursor which can support *A. macleodii* MIT1002 growth, and 3-methyl-2-oxopentanoic acid, the isoleucine precursor which cannot support *A. macleodii* MIT1002 growth (Fig. 1; structures provided in Fig. 4). These ketoacids are biochemically and structurally similar, and both are produced by *Prochlorococcus*, but 3-methyl-2-oxobutanoic acid is excreted at particularly high concentrations (Kujawinski et al. 2023). When cultured with 3-methyl-2-oxobutanoic acid, *A. macleodii* MIT1002 cellular abundance increased 94-fold over 59 h, and media concentrations of 3-methyl-2-oxobutanoic acid decreased eight-fold over the same period (Fig. 3A; Halloran et al. 2025). 3-Methyl-2-oxobutanoic acid uptake was modeled as a pseudo-first order process after normalizing for cell number, yielding a bulk uptake rate of $0.107 \mu\text{mol}$ 3-methyl-2-oxobutanoic acid $\text{cell}^{-1} \text{h}^{-1}$ (Supporting Information Fig. S9). In contrast, we observed no significant change in metabolite concentrations from the beginning to the end of the experiment in *A. macleodii* MIT1002 cultures incubated with 3-methyl-2-oxopentanoic acid (Fig. 3B; Welch's two sample t -test, $p = 0.53$). These cultures showed slight initial growth, presumably from residual carbon in the natural seawater, but cell abundance did not exceed that of controls without added metabolites (Fig. 3B).

Discussion

While *A. macleodii* is generally recognized to use diverse organic substrates (Noell et al. 2023; Pedler et al. 2014),

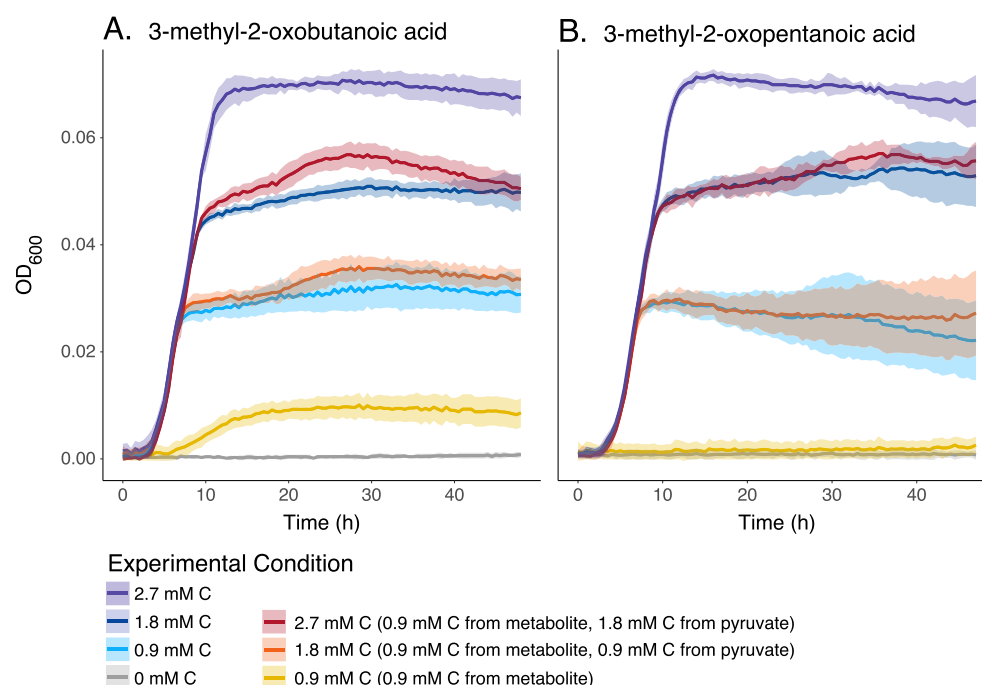


Fig. 1. *Alteromonas macleodii* MIT1002 growth on a selected *Prochlorococcus*-derived metabolite, pyruvate, or a mix of pyruvate and *Prochlorococcus* metabolite, across a range of carbon concentrations. Plot shows mean OD₆₀₀ ± 1 SD ($N \geq 3$). Selected metabolites shown here are (A) the valine intermediate, 3-methyl-2-oxobutanoic acid; and (B) the isoleucine intermediate, 3-methyl-2-oxopentanoic acid.

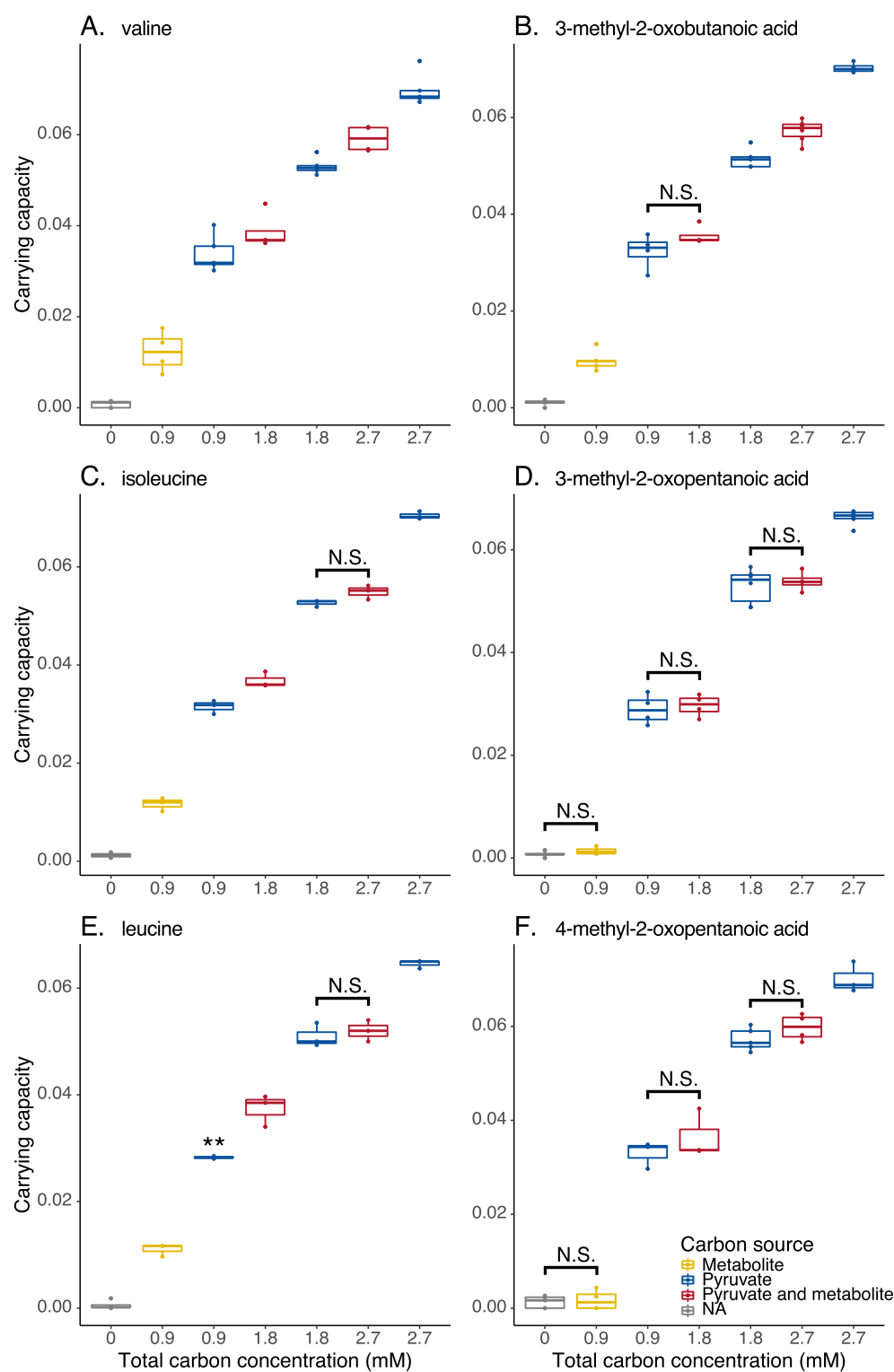


Fig. 2. Calculated carrying capacity of *A. macleodii* MIT1002 grown on a selected *Prochlorococcus* metabolite (yellow), pyruvate (blue), or a mix of pyruvate and the selected metabolite (red) across a range of carbon concentrations. Metabolites shown are (A) valine, (B) 3-methyl-2-oxobutanoic acid, (C) isoleucine, (D) 3-methyl-2-oxopentanoic acid, (E) leucine, and (F) 4-methyl-2-oxopentanoic acid. Carrying capacities under each condition were compared using a one-way ANOVA followed by post-hoc Tukey's HSD ($p < 0.05$). Conditions were significantly different except where marked N.S. (not significant). A double asterisk marks a condition where there were only 2 replicates.

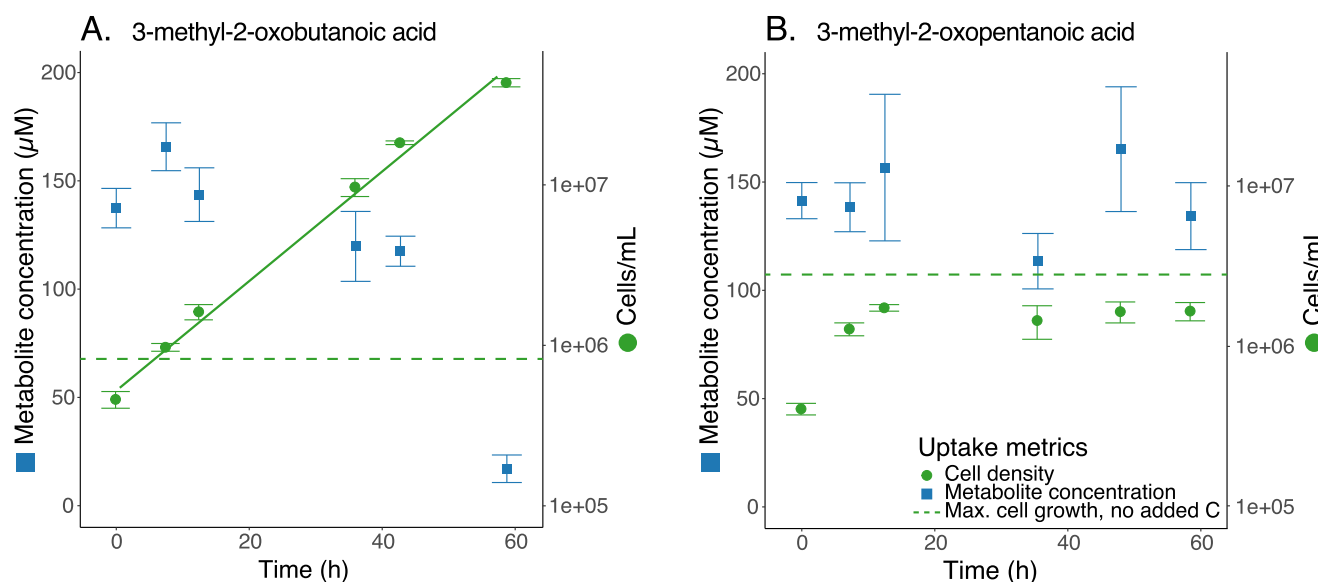


Fig. 3. Mean $\pm$ 1 SD ($N = 3$) of *A. macleodii* MIT1002 cell density (green circles) and extracellular metabolite concentration (blue squares) when incubated with either (A) 3-methyl-2-oxobutanoic acid or (B) 3-methyl-2-oxopentanoic acid. Dashed green line indicates maximum cell density reached in each experiment by negative controls (i.e., *A. macleodii* MIT1002 cultures incubated in seawater without metabolite addition). Solid line indicates the least squares regression fit to cell density over time for *A. macleodii* MIT1002 cultures grown with 3-methyl-2-oxobutanoic acid ($y = 5.74e^{0.03x}$, where y = cell density and x = time; $p < 0.05$, $R^2 = 0.99$).

studies have highlighted strain-specific genetic capabilities and preferences for specific compounds (Koch et al. 2020; Manck et al. 2020; Pedler Sherwood et al. 2015), consistent with our finding that *A. macleodii* MIT1002 only used some of the compounds tested (Table 1). In principle, substrate specificity could arise from intracellular mechanisms such as metabolite toxicity, where accumulation of a metabolite inhibits cell growth (Goldschmidt et al. 2018); hierarchical regulation (e.g., transcriptional or post translational regulation); or kinetic regulation, where enzyme activity controls metabolite flux (Millard et al. 2017). However, the varying response of *A. macleodii* MIT1002 to branched-chain keto acids is especially notable given their structural similarity, and their position in branched-chain amino acid metabolism. *Alteromonas macleodii* MIT1002 grows on all three branched-chain amino acids (valine, leucine, and isoleucine; Figs. 1, 2; Supporting Information Figs. S6–S8), indicating that it has complete catabolism pathways for these substrates. The first step in branched-chain amino acid catabolism is deamination to the respective branched-chain ketoacids via a branched-chain amino acid transferase (NCBI# MIT1002_00377; Fig. 4; Massey et al. 1976; Szentirmai and Horváth 1976). Therefore, *A. macleodii* MIT1002 can metabolize all three branched-chain ketoacids once they are present in the cell as part of branched-chain amino acid catabolism.

In the absence of intracellular regulation, transporter specificity likely underlies our observed substrate specificity. The degree of substrate specificity for 3-methyl-2-oxobutanoic acid

observed here is unexpected given previous studies of relevant transporters in marine microbes. A class of 2-oxoacid transporters common in marine microbes can transport the branched-chain keto acids, as well as unbranched 2-oxoacids like pyruvate and 2-oxopentanoic acid (Gonin et al. 2007; Pernil et al. 2010; Thomas et al. 2006). Indeed, both ligand binding assays with purified transporters and experiments with transporter mutants have characterized 2-oxoacid transporters which can bind all three of the branched-chain keto acids, albeit with slightly higher affinities for 3-methyl-2-oxobutanoic acid than for the other branched-chain keto acids (Pernil et al. 2010; Thomas et al. 2006). However, examination of the *A. macleodii* MIT1002 genome did not identify known 2-oxoacid transporter genes (Supporting Information Table S1), suggesting novel transporters are involved. Although previous studies have not reported a transporter which distinguishes between the branched chain keto acids, the promiscuity of transporters can vary widely even within a single transporter family. For example, one ABC transporter in *Bacillus subtilis* can transport carnitine or choline but not structurally similar osmolytes, while a closely related *B. subtilis* transporter can bind a wider range of osmolytes (Teichmann et al. 2017). Similarly, five point mutations in a myo-inositol transporter can abolish the transporter affinity for inositol (a 6-carbon sugar) and establish binding for ribose (a 5-carbon sugar) (Herrou and Crosson 2013).

Uptake experiments further support our hypothesis of transporter specificity. When cultured with 3-methyl-

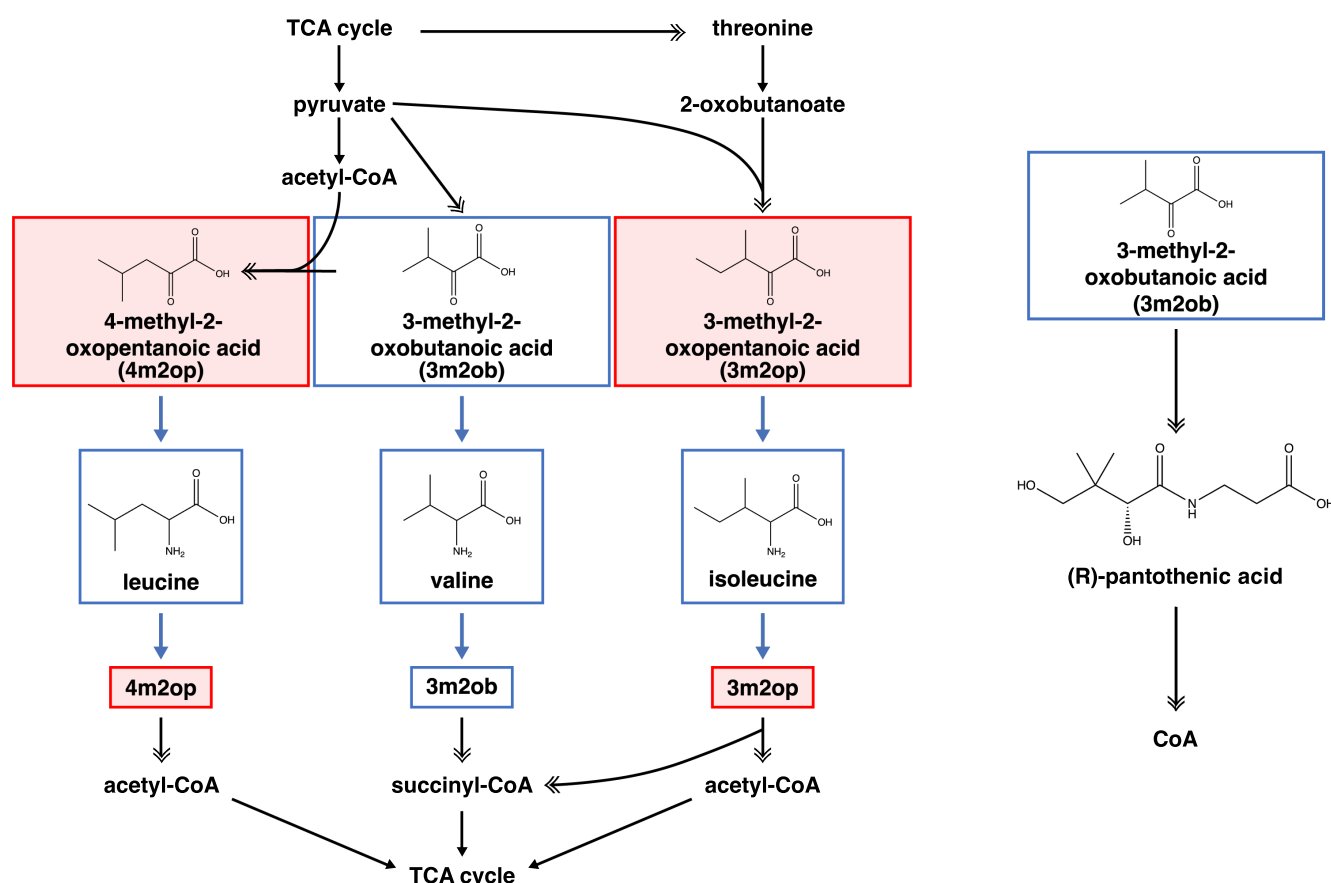


Fig. 4. Biosynthetic and catabolic pathways of the branched-chain amino acids (left) and of pantothenic acid (right). Metabolites which support *A. macleodii* strain MIT1002 growth are boxed in blue; metabolites which do not support *A. macleodii* MIT1002 growth are shaded and boxed in red. Thick blue arrows indicate a single shared enzyme (branched-chain amino acid transferase, NCBI# MIT1002_00377) which mediates transamination between the branched-chain amino acids and their immediate precursors/degradation products. Black arrows indicate enzymatic steps, with key intermediates listed; double headed arrows indicate two or more enzymatic reactions. TCA, tricarboxylic acid; CoA, coenzyme A.

2-oxobutanoic acid, *A. macleodii* MIT1002 increased in cellular abundance and removed the metabolite at a rate of $0.107 \mu\text{mol metabolite cell}^{-1} \text{h}^{-1}$, equivalent to $0.535 \mu\text{mol C cell}^{-1} \text{h}^{-1}$ (Fig. 3A; Supporting Information Fig. S9). In contrast, when cultured with 3-methyl-2-oxopentanoic acid, MIT1002 did not increase in cellular abundance beyond the control cultures, nor did it remove 3-methyl-2-oxopentanoic acid from solution (Fig. 3B). It is challenging to directly compare these rates to other reported compound-specific and/or microbe-specific uptake rates, due to differences in experimental design. For example, experiments designed to calculate Michaelis–Menten kinetic parameters found that bulk microbial communities from the North Pacific took up dissolved glycine betaine with a maximum uptake rate of $0.56 \text{ nmol L}^{-1} \text{h}^{-1}$ (Boysen et al. 2022), while SAR11 takes up glycine betaine with a maximum uptake rate of $14.92 \text{ nmol (mg cell)}^{-1} \text{h}^{-1}$ (Noell and Giovannoni 2019). In another study, incubations of *A. macleodii* strain AltSIO with bulk DOM found uptake rates of up to $0.06 \mu\text{mol C L}^{-1} \text{h}^{-1}$ (Pedler et al. 2014).

The specialization of *A. macleodii* MIT1002 on 3-methyl-2-oxobutanoic acid may reflect a more general role for this metabolite in marine microbial ecology. 3-Methyl-2-oxobutanoic acid was the most abundant branched-chain keto acid measured in the exudate of three *Prochlorococcus* ecotypes, and the only branched-chain keto acid produced by all three ecotypes (Kujawinski et al. 2023). Additionally, 3-methyl-2-oxobutanoic acid, but not the other branched-chain keto acids, has been measured in the dissolved phase in the oligotrophic Sargasso Sea, pointing to a possible broader environmental relevance for this metabolite (Longnecker et al. 2024). 3-Methyl-2-oxobutanoic acid may therefore be a prevalent substrate for microbial cross-feeding in the ocean. If so, microbes specializing on this metabolite could outcompete generalists who can take up a broader range of substrates. The observed substrate specificity of *A. macleodii* MIT1002 may therefore be an example of character displacement or niche differentiation, where a species specializes on a subset of available resources in the environment to minimize competition.

with other species (Pfennig and Pfennig 2010). Restricting the range of transporters expressed may also give an advantage to *A. macleodii* MIT1002 via viral resistance, as viruses often attach via transporters (Fuhrman 1999), and a general suite of transporters could expand the range of possible viral targets.

Biochemically, 3-methyl-2-oxobutanoic acid is a precursor in pantothenic acid biosynthesis (Fig. 4), which differentiates this metabolite from the other branched-chain keto acids. Pantothenic acid is the active component of coenzyme A, which is essential to all life and is used in central carbon metabolism and lipid biosynthesis (Leonardi and Jackowski 2007). The abundances of both intra- and extracellular pantothenic acid have distinct diel and seasonal patterns in the ocean, suggesting its importance as a marine metabolite (Boysen et al. 2021; Longnecker et al. 2024). The distinctive response of *A. macleodii* MIT1002 to 3-methyl-2-oxobutanoic acid, and the presence of this molecule in the field and in *Prochlorococcus* cultures, could in part reflect its dual role in the metabolism of branched-chain amino acids and of pantothenic acid.

For the *Prochlorococcus*-derived metabolites that could support *A. macleodii* MIT1002 growth, we observed diauxic growth curves in cultures which received a mix of pyruvate and metabolite (Fig. 1; Supporting Information Figs. S6–S8). This biphasic growth pattern suggests that *A. macleodii* MIT1002 uses substrates sequentially, with pyruvate likely used first. This prioritization effect could be due to the higher growth rate and/or higher carrying capacity when cells were grown on pyruvate relative to other substrates (Fig. 1; Supporting Information Figs. S6–S8). This is consistent with similar examples of substrate prioritization based on growth rate and/or carrying capacity in other systems (Perrin et al. 2020). The lower carrying capacity from growth on substrates other than pyruvate is seen both in the comparison of cultures grown on a single substrate and when a portion of carbon from pyruvate was replaced with the other substrates (Fig. 2). Differences in carrying capacity between substrates reflects differences in their bacterial growth efficiency (BGE), which is the fraction of carbon in substrates assimilated into biomass. Variations in BGE between substrates has been observed elsewhere and can occur as a function of the oxidation state of the substrate, enzymatic steps required for substrate catabolism, energy requirements for substrate transport and metabolism, or genetic capacity of the microbe (Bölscher et al. 2016; Brown and Jones 2024; del Giorgio and Cole 1998; Frey et al. 2013; Manzoni et al. 2012; Öquist et al. 2017; Pold et al. 2020; Qiao et al. 2019; Roller and Schmidt 2015). In the case of *A. macleodii* MIT1002, we hypothesize that the difference in BGE between substrates is due to differences in enzymatic requirements of associated pathways. That is, while pyruvate can be converted to oxaloacetic acid or acetyl CoA, which feed directly into the citric acid cycle (Voet and Voet 2011), the catabolism of branched-chain amino acids or 3-methyl-2-oxobutanoic acid into central carbon metabolism

requires many more enzymatic steps through branched-chain coenzyme A intermediates (Massey et al. 1976). This may lead to higher resource or energy investments in using these compounds, potentially lowering the BGE.

Conclusion

The interactions between microbes and metabolites are crucial to both microbial ecology and carbon cycling. Here we show that *A. macleodii* MIT1002 grows selectively on *Prochlorococcus*-derived metabolites, likely via transporter specificity. These results further expand our knowledge of the metabolic interactions between *Prochlorococcus* and *Alteromonas*, and offer 3-methyl-2-oxobutanoic acid as a potentially important molecular mediator of these interactions. The specificity of the *A. macleodii* MIT1002 response demonstrates the importance of characterizing microbe-molecule interactions and dissolved organic matter with compound-specific resolution. Small changes in molecular structure can have an outsized effect on microbial response, and molecular-level studies enable a nuanced understanding of this interplay that would not be possible through bulk characterization alone.

Author Contributions

Kathryn H. Halloran, Sallie W. Chisholm, and Elizabeth B. Kujawinski conceptualized the experiments. Kathryn H. Halloran carried out the experiments, along with Allison Coe and Gretchen Swarr. Melissa C. Kido Soule led the UHPLC–MS/MS analysis. Kathryn H. Halloran, Rogier Braakman, Allison Coe, and Elizabeth B. Kujawinski undertook statistical analysis. Kathryn H. Halloran wrote the original draft, and all authors edited and reviewed subsequent drafts. Sallie W. Chisholm, Elizabeth B. Kujawinski, and Rogier Braakman secured funding.

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Conflicts of Interest

None declared.

Data Availability Statement

LC-MS/MS data have been submitted to MetaboLights under the identifier MTBLS10981. Data and metadata from all experiments are available through the Environmental Data Initiative (EDI) data portal at <https://doi.org/10.6073/pasta/e74a8ce04ccc317c3c3208f3c382f0a2>. Data from all experiments, and the code used to analyze those data, can also be found at http://github.com/khhalloran/Alteromonas_mtab_uptake.

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Supporting Information

Additional Supporting Information may be found in the online version of this article.

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